M8-VIDEO

Video Card for the Mark-8 Minicomputer

User's Guide

Rev B PC Boards

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4 October 2026

Revision History

Manual	PCB	Date	Author	Notes

TABLE OF CONTENTS

Overview 4
Theory of Operation and Design Philosophy 4
Assembly and Test Instructions 5
General Assembly Tips 5
Assembly 5
Testing 11
PC Board Components 14
Appendix A: Software 17
PC Board Drawings 20
Schematic Drawings 22

OVERVIEW

The M8-VIDEO PCB is a video card designed by Len Bayles for the Mark-8 minicomputer.

The card provides the following capabilities:

1. Accepts 8-bit data from a Mark-8 output port and displays the corresponding character on a composite-video output.
2. Automatically advances the cursor for each character.
3. Accepts a manual cursor position by writing the COL and ROW positions via separate output ports.
4. Supports a 128-character set stored on a 2716 EPROM.
5. Displays up to 512 characters on the screen.

Theory of Operation and Design Philosophy

The display is a 32-column by 16-row character grid (512 cells).

This PCB contains 3 configurable ports for data, column position and row position. Rev B has these ports pre-configured as follows: Character on port 5, column position on port 6 and row position on port 7. The position is auto-incremented after each character is written but can be set to any position by writing to the row/col output ports.

To reset the cursor to position 0 (top left), the user must write 0 to both the column and row output ports. To clear the screen, the user must write a SPACE to all positions on the screen (all 512 cells).

Character generation is performed by a 2716 EPROM (128 characters). This allows the user to customize the character-set as needed. A full character-set Intel Hex file is included in Appendix A. Seven 2102 SRAM devices store the current characters that are displayed on the screen.

This board is powered by +5V only and contains circuitry for converting +5V to -12V for the sync generator.

ASSEMBLY AND TEST INSTRUCTIONS

General Assembly Tips

Install components in order of increasing height, beginning with the lowest-profile parts. This allows the work surface to hold the parts against the board when the assembly is inverted for soldering.

Refer to the PC Board Drawings (page 18) for the location of each component.

Install all components of a given type before proceeding to the next type (for example, all resistors, then all IC sockets). Secure the parts with low-adhesion tape (painter's tape) before inverting the board for soldering.

When installing DIP sockets, seat each device fully against the board. Solder two diagonally opposite pins first. After all such devices are tacked in place, solder the remaining pins.

Warning: Some traces are very close to the solder pads. Use a small tip when soldering and be very careful to prevent solder shorts.

Assembly

Step 1 Install resistors

Variable resistors will be installed later. Notice that there are 4 x R101 resistors - they are all 1K Ω.

Component	Value	Location	Quantity
Resistor, 1/4W	330 Ω	R1-R2	2
Resistor, 1/4W	8.2K Ω	R3	1
Resistor, 1/4W	22K Ω	R5,R15	2
Resistor, 1/4W	5.1K Ω	R7	1
Resistor, 1/4W	1K Ω	R8,R101(4)	5
Resistor, 1/4W	2.2K Ω	R9	1
Resistor, 1/4W	56 Ω	R10	1
Resistor, 1/4W	68 Ω	R11	1
Resistor, 1/4W	5K Ω	R12-R13	2
Resistor, 1/4W	3.9K Ω	R14	1

Step 2 Install inductor

Component	Value	Location	Quantity
Inductor, Axial	6.8 uH, >200 mA	L1	1

Step 3 Install DIP sockets

Component	Location	Quantity
8-pin DIP socket, 0.3" wide	U40	1
14-pin DIP socket, 0.3" wide	U3-U4,U6,U10,U16-U17,U21,U38-U39,U41	10
16-pin DIP socket, 0.3" wide	U2,U5,U7-U9,U11-U13,U15,U18-U20,U22,U24-U37	27
20-pin DIP socket, 0.3" wide	U23	1
24-pin DIP socket, 0.6" wide	U14	1

See General Assembly Tips above. Tack each socket with two diagonally opposite pins, and complete soldering after the DIP switch has been similarly tacked in place.

Step 4 Install transistors

Component	Value	Location	Quantity
Transistor, NPN, TO-92	2N2222	Q1-Q2	2

Step 5 Install crystal

Component	Value	Location	Quantity
Crystal	14.31818 MHz	X1	1

Bend the leads of the crystal at a 90 degree angle so that the crystal could lay flat on the PCB. Solder a piece of wire to the crystal body and the ground plane on the underside of the PCB where the crystal lies.

Step 6 Install ceramic capacitors

Component	Value	Location	Quantity
Capacitor, Ceramic	0.01 uF	C1,C10	2
Capacitor, Trimmer	50 pF	C2	1
Capacitor, Ceramic, C0G*	220 pF	C3	1
Capacitor, Ceramic	1000 pF	C5	1
Capacitor, Ceramic	0.22 uF	C6	1
Capacitor, Ceramic	100 pF	C7	1
Capacitor, Ceramic	0.05 uF	C8	1
Capacitor, Ceramic	0.1 uF	C101-C106,C201-C206, C301-C306,C401-C406	24

* Using a Y5P or similar may cause the clock to be irregular and the video jittery if the voltage changes often. Use a C0G or a silver mica for a cleaner clock signal.

Step 7 Install electrolytic capacitors

C11 is the electrolytic capacitor that is not marked on the PCB. It is located near pin 3 and pin 6 of the bottom connector.

Component	Value	Location	Quantity
Capacitor, electrolytic, 25 V, axial	4.7 uF	C4	1
Capacitor, electrolytic, 25 V, radial	10 uF	C9	1
Capacitor, electrolytic, 25 V, axial	100 uF	C11	1

Observe polarity. The positive lead is marked “+” on the PCB.

Step 8 Install connectors

Seat each connector flush with the board. Solder one pin at each end, verify alignment, then solder the remaining pins.

J1 is not a standard single-piece 41-position connector. Assemble it from standard Molex 0.156 in housings; for example, three 10-position connectors and one 11-position connector.

Component	Value	Location	Quantity
Right-angle female connector, Molex 0.156 in, 41-position		J1	1
RCA Jack	Same Sky #RCJ-014	J2	1

Step 9 Install DC converter

Component	Value	Location	Quantity
DC Converter, +5V In, -12V Out, 2W	102-2732-ND	PS1	1

Step 10 Install variable resistors

Component	Value	Location	Quantity
Resistor, Variable, 1/2W	50K Ω	R4,R6	2

Step 11 Install jumpers (optional)

Rev B has been prewired for output ports 5-7. If you wish to change this configuration, refer to the section titled “Step 15 Configuration”.

Step 12 Inspect the assembly

1. Inspect all solder joints for omissions and bridges.
2. With an ohmmeter, check for a short between J10 pin 3 (GND) and pin 6 (+5 V). Installed components produce a finite resistance; a near-zero reading indicates a short circuit, which must be corrected before proceeding.

Step 13 Clean the board

If organic-core or water-soluble-flux solder was used, cleaning is required. To remove water-soluble flux:

1. Wash the board after soldering is complete and before ICs are inserted. Hold the board under running hot water and scrub with a brush for several minutes.
2. Remove water with compressed air, including from beneath components and from inside IC sockets. A workshop compressor is more effective than aerosol dusters.
3. Dry the board near a moderate heat source for approximately 30 minutes.

If rosin-core solder was used, cleaning is optional. Apply a flux remover suitable for rosin; a brush may be used to loosen residue. Inspect the solder joints again after cleaning.

Step 14 Install ICs

U14 (2716 EPROM) needs to be programmed before use. Refer to the section titled "Step 15 Configuration" for instructions on how to do that before continuing.

Insert each IC with pin 1 toward the top of the board (except for U14 which has pin 1 in the top right), as shown on the PC Board Drawings. Do not fold pins under the body. If necessary, press each row of pins against the work surface to set the lead pitch before insertion.

Refer to the PC Board Drawings for IC locations and orientations.

Component	Value	Location	Quantity
IC, DIP, Hex Inverter	7404	U1,U16	2
IC, DIP, Dual JK Flip-Flop, Set/Clear	74LS76	U2	1
IC, DIP, Counter, Divide-by-Twelve	7490	U3	1
IC, DIP, Quad 2-Input NAND Gate	7400	U4	1
IC, DIP, Sync Generator	MM5321	U5	1
IC, DIP, Triple 3-Input NAND Gate	7410	U6,U39	2
IC, DIP, 4-Bit Binary Counter	74193	U7-U9,U11-U12, U18-U20	8
IC, DIP, Quad 2-Input NAND Gate	74LS00	U10	1
IC, DIP, Dual Monostable Multivibrator	74LS123	U13,U34	2
IC, DIP, EPROM	2716	U14	1
IC, DIP, 8-Bit Shift Register	74165	U15	1

IC, DIP, 8-Input NAND Gate	7430	U17	1
IC, DIP, Quad 2-Input OR Gate	74LS32	U21	1
IC, DIP, 3:8 Line Decoder	74LS138	U22	1
IC, DIP, Octal Buffer/Line Driver	74LS244	U23	1
IC, DIP, Quad 2:1 Multiplexer	74157	U24-U26	3
IC, DIP, 1-KBit SRAM	2102	U27-U33	7
IC, DIP, 4-Bit Magnitude Comparator	74LS85	U35-U37	3
IC, DIP, Dual D Flip-Flop	74H74	U38	1
IC, DIP, Timer	555	U40	1
IC, DIP, Hex Inverter	74LS04	U41	1

Step 15 Configuration

A. EPROM Programming

U14 (2716 EPROM) acts as the character generator for the video card. The assembler can program the EPROM with any character-set desired but if a standard ASCII character set is desired, program the EPROM with the “Character Set Intel Hex” located under Appendix A as that character-set contains all ASCII characters.

The basic idea behind the EPROM character generation is that each character address is mapped to A0-A6 of the EPROM. The remaining address lines (A7-A10) are the row-select for that character. For example, the ASCII character “T” is binary 01010100. A0 is 0, A1 is 0, A2 is 1, and so on. Address lines A7-A10 would determine what is printed for that specific row for the ASCII character “T”. When A7-A10 is “0000”, the value at that location will be 7F or 01111111 in binary. That is the top bar of the letter “T”. When A7-A10 is “0001”, the value at that location will be 08 or “00001000” in binary. That is the first part of the vertical bar of the letter “T”. Continuing in this matter will generate all rows of the letter “T”.

An EPROM programmer that works well with the 2716 is XGecu T48 [TL866-3G].

B. Port Selection

Rev B has output ports 5-7 pre-wired. If the assembler desires a different port for any of the ports used (DI, CA, RA), he can cut the small traces for each of the address pads (marked as DI, CA and RA on the PCB) on the underside of the

PCB and then solder jumper wires to the desired address on the front of the PCB (marked as 0-7). *Note that the traces to cut are smaller than the standard traces.*

For example, if the user desired port 3 to be used for the data, he can cut the trace for DI on the underside of the PCB (currently going to "5") and solder a jumper wire between DI and "3".

If the user desires an address higher than 0-7 (i.e. port 16), the trace for "I" (in) needs to be cut and tied to the "A13" pad (left one). The "A13" trace will also need to be cut and tied to the "O" (out) pad. The "DI" pad will need to be tied to "0". This setup can be used to address ports 16-23.

For ports 8-15, follow the same procedure but for the "A12" pad instead of the "A13" pad.

Step 16 — Final inspection

Inspect the completed board for the following defects:

- IC installed with reversed orientation
- IC pin folded under the package
- Incorrect device in any socket
- ICs not fully seated

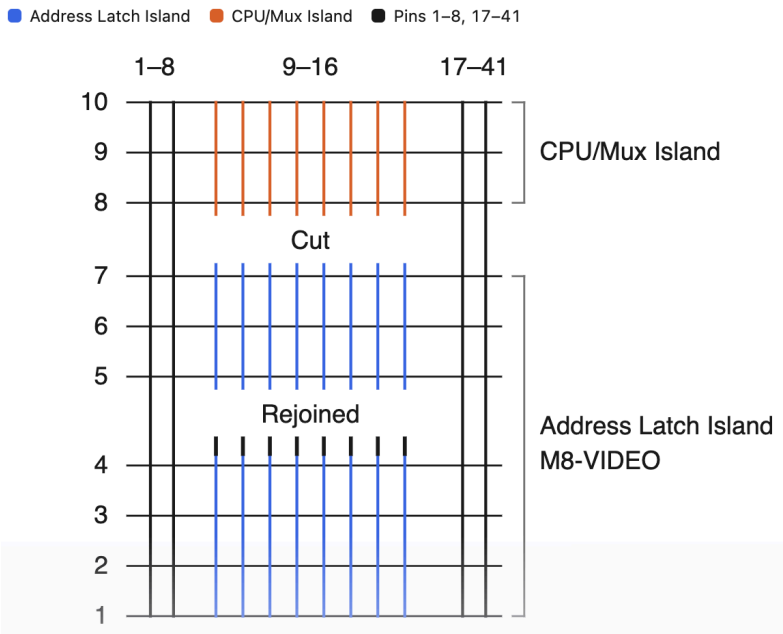
Testing

The following procedure verifies basic operation of the M8-VIDEO in a working Mark-8 system. It is not a complete troubleshooting guide. Diagnosis of a non-functional system is beyond the scope of this manual.

Step 1 Backplane preparation

The M8-VIDEO PCB needs to be inserted in the Mark-8 backplane where the 8 data lines on the backplane are outside the CPU/Input Multiplexer input cycle. What this means is that, most likely, you will need to tie the originally cut data lines (pins 9-16) on the backplane back together in order to have a slot open for the M8-VIDEO PCB. And then cut the same traces (pins 9-16) further down the backplane so that only the CPU board, Input Multiplexer board and any other board that might share the same cycle are tied together on those lines. All the other boards need to be on a separate island. It's recommended to leave room for at least 3 boards at the back of the backplane for the CPU board, multiplexer board and one additional board.

Refer to the following diagram of the backplane changes:



Step 2 Installation

Insert the M8-VIDEO PCB into the Mark-8 backplane in one of the "Address Latch Island" slots so that data pins 9-16 do not interfere with the CPU board.

Connect a composite-video monitor to the RCA jack of the M8-VIDEO PCB.

Step 3 Video test

Power on the computer. The operator should see a video screen filled with garbage data. This is normal. Enter the following program at memory address 0:

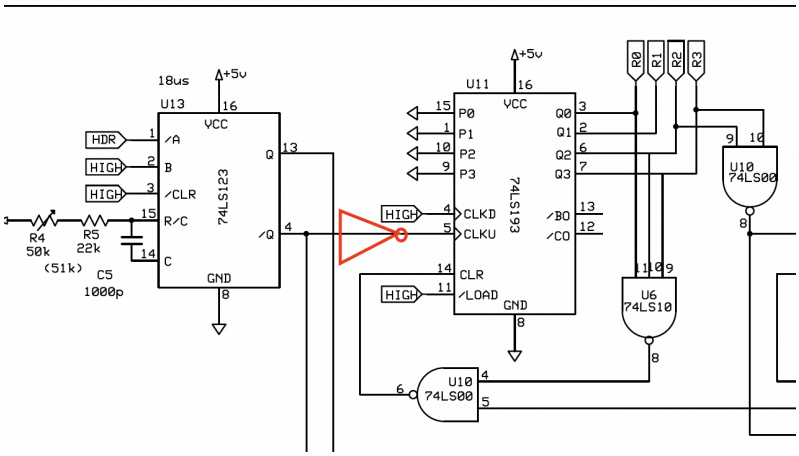
Address	Machine Code	Assembly Code
0	006 000	LAI 000 / Load '0' into A register
2	135	OUT 6 / Set COL to 0
3	137	OUT 7 / Set ROW to 0
4	006 201	LAI 201 / Load 'A' into A register
6	133	OUT 5 / Output A register to port 5
7	377	HLT

Execute from address 0 (interrupt with 105 octal). After the processor halts, the first character on the screen should be the letter 'A'.

Issues

Issue 1 First row of dots missing

If you have an issue where the first row of dots is not shown, try modifying the PCB to invert the signal going into U11 pin 5. U16 has some unused inverters that can be used for this purpose.



PC BOARD COMPONENTS

Component	Value	Location	Quantity
PC Board, M8-VIDEO, Rev B			1
Resistor, 1/4W	330 Ω	R1-R2	2
Resistor, 1/4W	8.2K Ω	R3	1
Resistor, 1/4W	22K Ω	R5,R15	2
Resistor, 1/4W	5.1K Ω	R7	1
Resistor, 1/4W	1K Ω	R8,R101(4)	5
Resistor, 1/4W	2.2K Ω	R9	1
Resistor, 1/4W	56 Ω	R10	1
Resistor, 1/4W	68 Ω	R11	1
Resistor, 1/4W	5K Ω	R12-R13	2
Resistor, 1/4W	3.9K Ω	R14	1
Inductor, Axial	6.8 uH, >200 mA	L1	1
8-pin DIP socket, 0.3" wide		U40	1
14-pin DIP socket, 0.3" wide		U3-U4,U6,U10,U16-U17,U21,U38-U39,U41	10
16-pin DIP socket, 0.3" wide		U2,U5,U7-U9,U11-U13,U15,U18-U20,U22,U24-U37	27
20-pin DIP socket, 0.3" wide		U23	1
Transistor, NPN, TO-92	2N2222	Q1-Q2	2
Crystal	14.31818 MHz	X1	1
Capacitor, Ceramic	0.01 uF	C1,C10	2
Capacitor, Trimmer	50 pF	C2	1
Capacitor, Ceramic, C0G	220 pF	C3	1
Capacitor, Ceramic	1000 pF	C5	1

Capacitor, Ceramic	0.22 uF	C6	1
Capacitor, Ceramic	100 pF	C7	1
Capacitor, Ceramic	0.05 uF	C8	1
Capacitor, Ceramic	0.1 uF	C101-C106,C201-C206,C301-C306,C401-C406	24
Capacitor, electrolytic, 25 V, axial	4.7 uF	C4	1
Capacitor, electrolytic, 25 V, radial	10 uF	C9	1
Capacitor, electrolytic, 25 V, axial	100 uF	C11	1
Right-angle female connector, Molex 0.156 in, 41-position		J1	1
RCA Jack	Same Sky #RCJ-014	J2	1
DC Converter, +5V In, -12V Out, 2W	102-2732-ND	PS1	1
Resistor, Variable, 1/2W	50K Ω	R4,R6	2
IC, DIP, Hex Inverter	7404	U1,U16	2
IC, DIP, Dual JK Flip-Flop, Set/Clear	74LS76	U2	1
IC, DIP, Counter, Divide-by-Twelve	7490	U3	1
IC, DIP, Quad 2-Input NAND Gate	7400	U4	1
IC, DIP, Sync Generator	MM5321	U5	1
IC, DIP, Triple 3-Input NAND Gate	7410	U6,U39	2
IC, DIP, 4-Bit Binary Counter	74193	U7-U9,U11-U12,U18-U20	8
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IC, DIP, EPROM	2716	U14	1
IC, DIP, 8-Bit Shift Register	74165	U15	1

IC, DIP, 8-Input NAND Gate	7430	U17	1
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IC, DIP, 3:8 Line Decoder	74LS138	U22	1
IC, DIP, Octal Buffer/Line Driver	74LS244	U23	1
IC, DIP, Quad 2:1 Multiplexer	74157	U24-U26	3
IC, DIP, 1-KBit SRAM	2102	U27-U33	7
IC, DIP, 4-Bit Magnitude Comparator	74LS85	U35-U37	3
IC, DIP, Dual D Flip-Flop	74H74	U38	1
IC, DIP, Timer	555	U40	1
IC, DIP, Hex Inverter	74LS04	U41	1

APPENDIX A: SOFTWARE

Character Set Intel Hex

Intel Hex that includes all ASCII characters.

```
:10000000000000300016001800004000000800004A
:10001000000000001000000001F000008007F002A
:10002000000812140820382008080800000000000A
:100030003E083E3E027F3E7F3E3E00000400103EF2
:100040003E1C7E1E7E7F7F1E413E1F414041411C63
:100050007E1C7E3E7F41414141417F1E003C3E006F
:10006000020040000200C00400000410000000B0
:10007000000000000000000000000000E0838307F83
:10008000000000048000E00240000200000 1C0000BA
:10009000000000001000000001C1000001C002030B8
:1000A000000812143F5144201004490800000001C8
:1000B000411841410640414141410000080008 41CA
:1000C000412221212140402141080442406361 2214
:1000D000412241410841414141410110400441 0058
:1000E000020040000200120040080040100000 0022
:1000F000000000001000000000000100804497F0C
:10010000000000401810004200401000002000 0D5
:10011000000000003800000022100410 2A081049D6
:10012000000812144822444020022A08000000025D
:10013000430801010A4040024141200010000441EF
:10014000414121402140404041080444405551 4133
:10015000414141400841414122220210200400 0057
:10016000010040000200100040000040100000 00AC
:10017000000000001000000000000001008040 67FCE
:10018000003C004020200042404810006218183F08
:10019000181F3F6254000822411002200800080680
:1001A0000008007F4804280020021C080000000040A
:1001B000450802011240400441410020207F020214
:1001C0004D41214021404040410804484049494 1B7
:1001D000414141400841414914140410100400 00F9
:1001E000001E5C3C3A3C103A5818024410765C3CC5
:1001F0005C3A5C3C7C42444142427E100004007FF9
:10020000312261204040 2C7E4050104822202454 4E
:10021000242448245462494141107F7F087F0630DE
:10022000000800143E08100020027F7F007F0008B5
:1002300049080C1E227E7E083E3F0000400001045B
:10024000557F3E40217878407F0804704049454101
:100250007E417E3E0841414908080810080400001C
:10026000002262424642 7C46640802481049624 2CB
:10027000624662421042444924420420000200 7F48
:100280004A3C12307840524240601048244042 14A8
:10029000424208245414 2A4941100220080008 4907
:1002A0000008007F0910290020021C08000000102F
:1002B000510810017F01411041010000207F020818
:1002C0005E41214021404047410804484041434 1AC
:1002D000404148010841414914081010040400 003D
:1002E000004242404242104242080270104942 42DB
:1002F000424240301042444918420810080400 7F2E
:10030000442214484038124240501848283C4214B5
:10031000424208243808 2A49225004100808100 6CE
:10032000000000140922460020022A0800000020D4
:10033000610820010201411041012000100004 0861
```

```
:1003400040412140214040414108044440414141 55
:1003500040454401084122552208201002040000B3
:1003600000424240427E1042420802481049424246
:1003700042424 00C1042444918421010080400 7FC9
:100380004A2218482004122448 4A244830022414DF
:1003900064240824101 42A4922300000800200098
:1003A000000000147E4546001004490800000040 8B
:1003B000410840410241411041410000800080 04D
:1003C000414121212140402141084442404141 22F4
:1003D000402242410841146341084010010400 00DA
:1003E000046624246401046420802441049424 2DA
:1003F000624640421242284924462010080400 7FE9
:10040000313C1030181812183044427420 0C181463
:1004100058180818102 31C366310000008007F00CD
:1004200000080014080239000808080030001000 15
:100430003E3E7F3E023E3E103E3E0030040010082D
:100440003E417E1E7E7F401E413E38417F41411C21
:10045000401D413E083E084141087F1E003C007F90
:10046000003A5C3C3A3E103A420802421049423C93
:100470005C3A403C0C3C1036423A7E0E0038007F1D
:1004800000203000000002000000004000000000DA
:1004900040000000000008000000000000000000 14
:1004A000000000000000000000000000010000000 3C
:1004B000000000000000000000000010000000002C
:1004C000000000000000000000000000000000002C
:1004D000000000000000000000000000000000001C
:1004E000000000000000020000200000000000 08
:1004F000400200000000000000200000000000B8
:1005000000203000000002000000004000000000 59
:1005100040000000000008000000000000000000 93
:1005200000000000000000000000000010000000BB
:1005300000000000000000000000010000000000AB
:1005400000000000000000000000000000000000AB
:1005500000000000000000000000000000000009B
:1005600000000000000004200004200000000000 07
:1005700040020000000000000420000000000F7
:1005800000403000000002000000004000000000B9
:1005900040000000000008000000000000000000 13
:1005A000000000000000000000000002000000002B
:1005B000000000000000000000000020000000001B
:1005C000000000000000000000000000000000002B
:1005D000000000000000000000000000000000001B
:1005E000000000000000003C00003C000000000093
:1005F000400200000000000003C0000000000007D
:10060000000000000000000000000000000000 0EA
:10061000000000000000000000000000000000 0DA
:10062000000000000000000000000000000000 0CA
:10063000000000000000000000000000000000 0BA
:10064000000000000000000000000000000000 0AA
:10065000000000000000000000000000000000 09A
:10066000000000000000000000000000000000 08A
:10067000000000000000000000000000000000 07A
:10068000000000000000000000000000000000 06A
:10069000000000000000000000000000000000 05A
:1006A000000000000000000000000000000000 04A
:1006B000000000000000000000000000000000 03A
:1006C000000000000000000000000000000000 02A
:1006D000000000000000000000000000000000 01A
:1006E000000000000000000000000000000000 00A
:1006F000000000000000000000000000000000 0FA
```

```
:10070000000000000000000000000000000000000000E9
:10071000000000000000000000000000000000000000D9
:10072000000000000000000000000000000000000000C9
:10073000000000000000000000000000000000000000B9
:10074000000000000000000000000000000000000000A9
:10075000000000000000000000000000000000000000099
:10076000000000000000000000000000000000000000089
:10077000000000000000000000000000000000000000079
:10078000000000000000000000000000000000000000069
:10079000000000000000000000000000000000000000059
:1007A000000000000000000000000000000000000000049
:1007B000000000000000000000000000000000000000039
:1007C000000000000000000000000000000000000000029
:1007D000000000000000000000000000000000000000019
:1007E000000000000000000000000000000000000000009
:1007F000000000000000000000000000000000000000F9
:00000001FF
```

"Hi" Output

This small piece of code writes "Hi" to the screen at position 0.

```
;
; tvt.asm
;
; Intel 8008 source code file.
;
; Copyright 2015 by Len Bayles
;

CHAR_OUT:     equ 5
COL_OUT:      equ 6
ROW_OUT:      equ 7


; page 0

       cpu    8008

       org    000#000

main:
       lai    0x00
       out    COL_OUT       ; Put value on output port
       out    ROW_OUT       ; Put value on output port

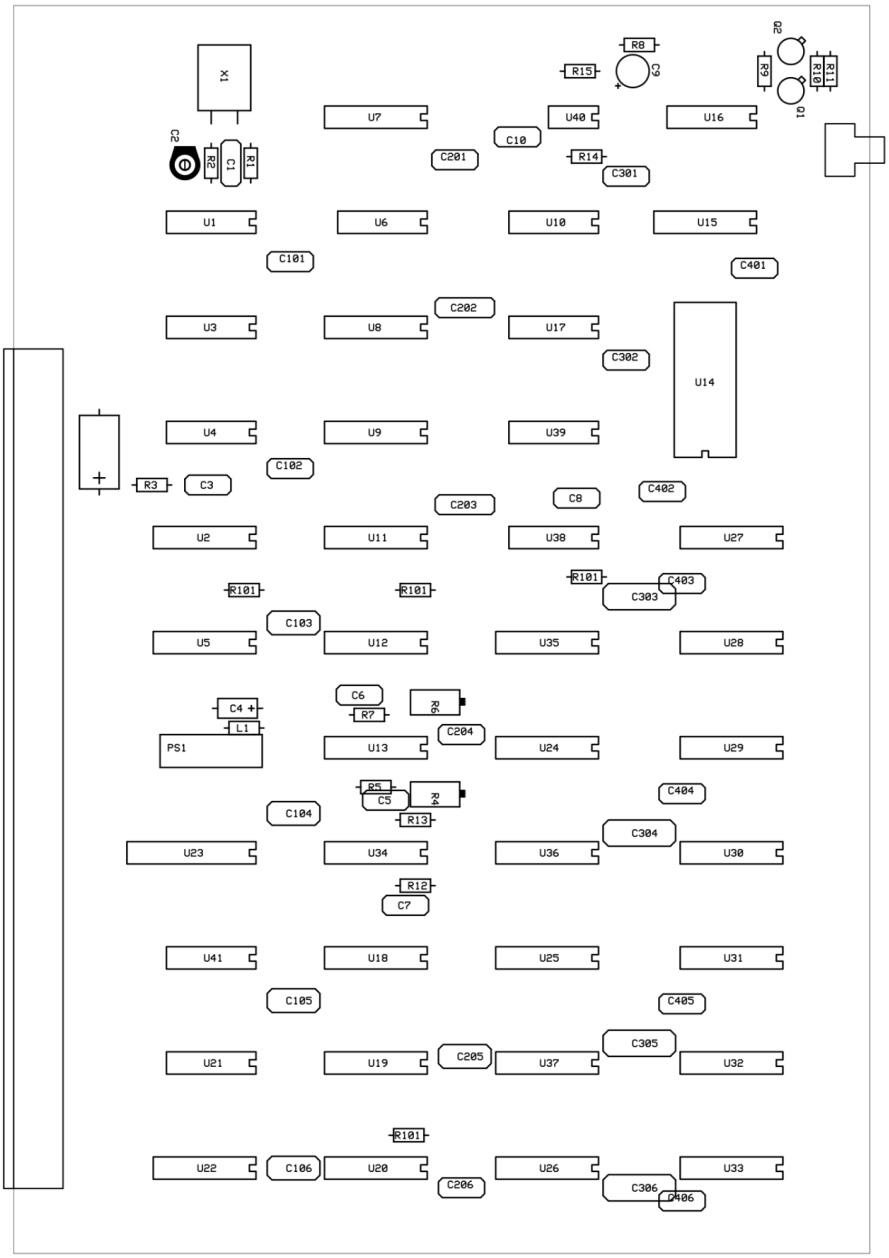
       lai    0x48
       out    CHAR_OUT      ; Put value on output port

       lai    0x69
       out    CHAR_OUT      ; Put value on output port


       end
```

PC BOARD DRAWINGS

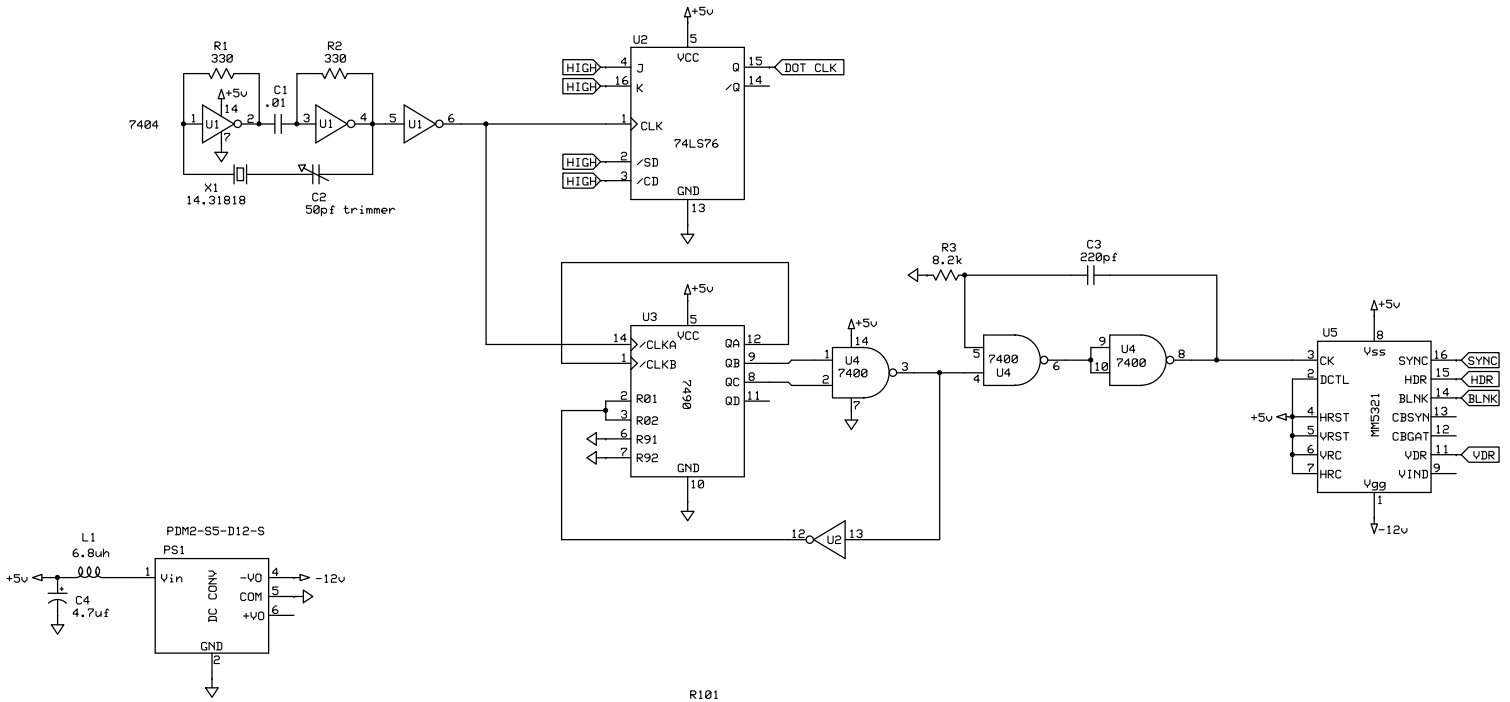
The following pages show component placement for Rev B.



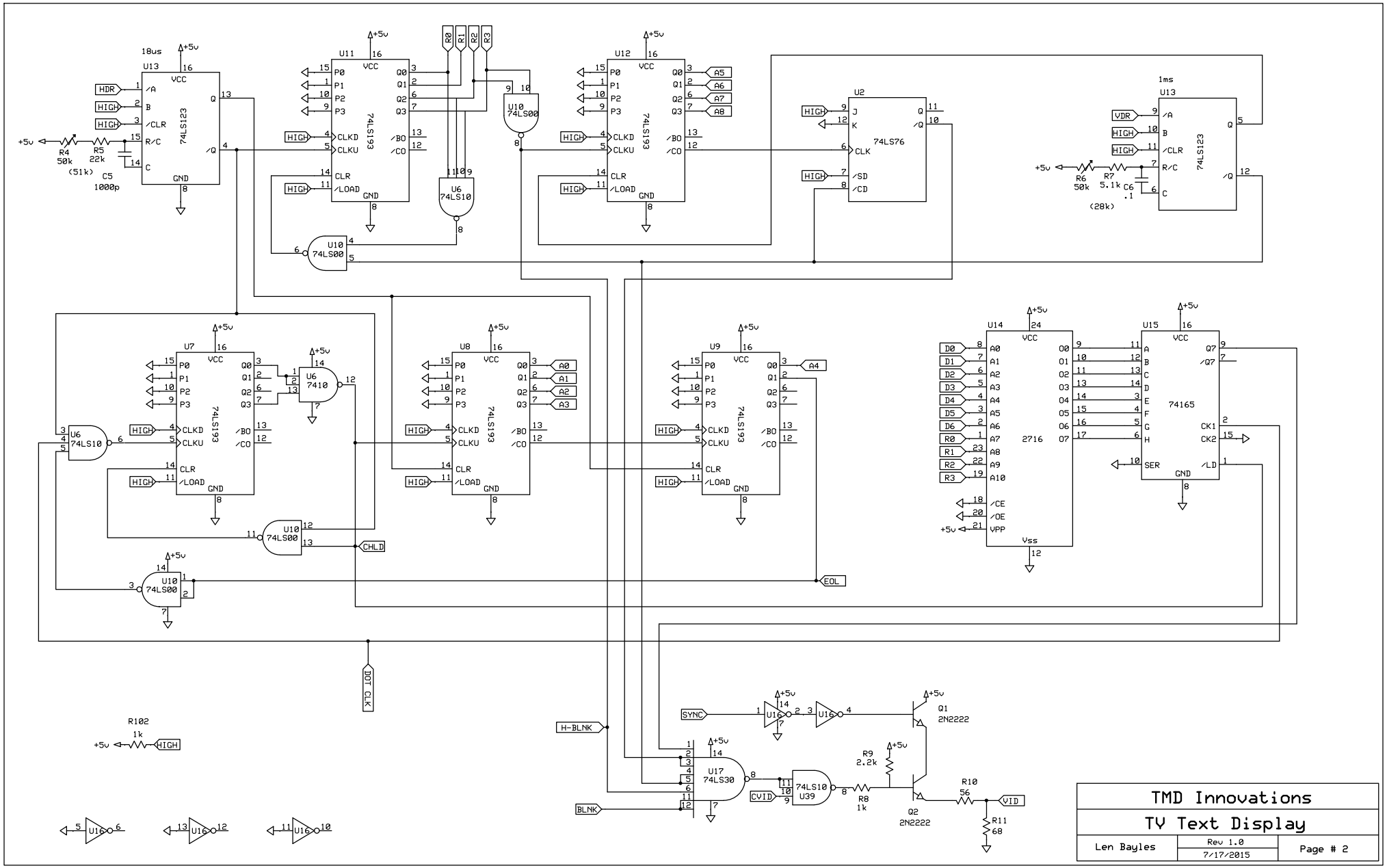
Rev B Component Placement

SCHEMATIC DRAWINGS

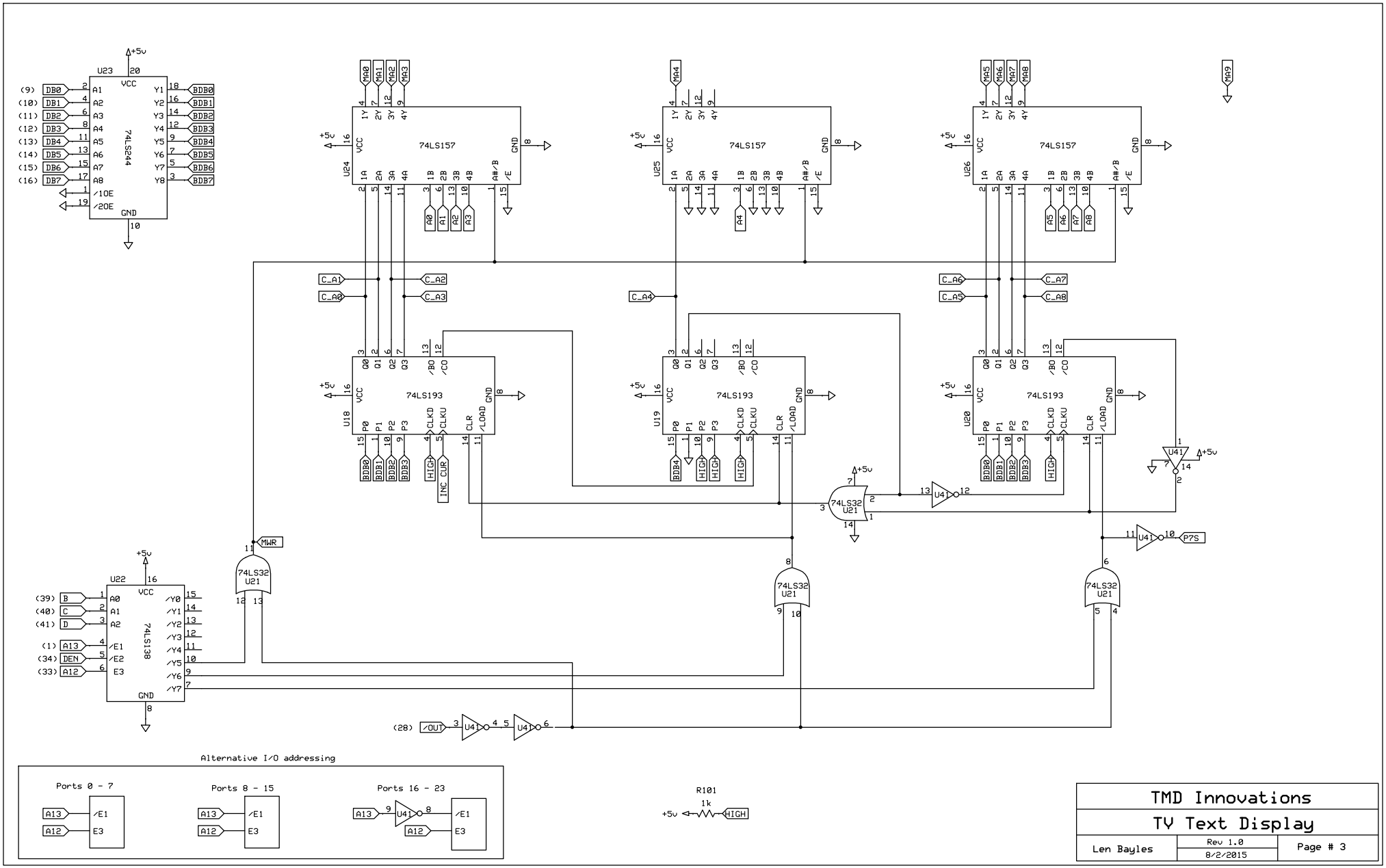
The following pages contain the schematic for Rev B.



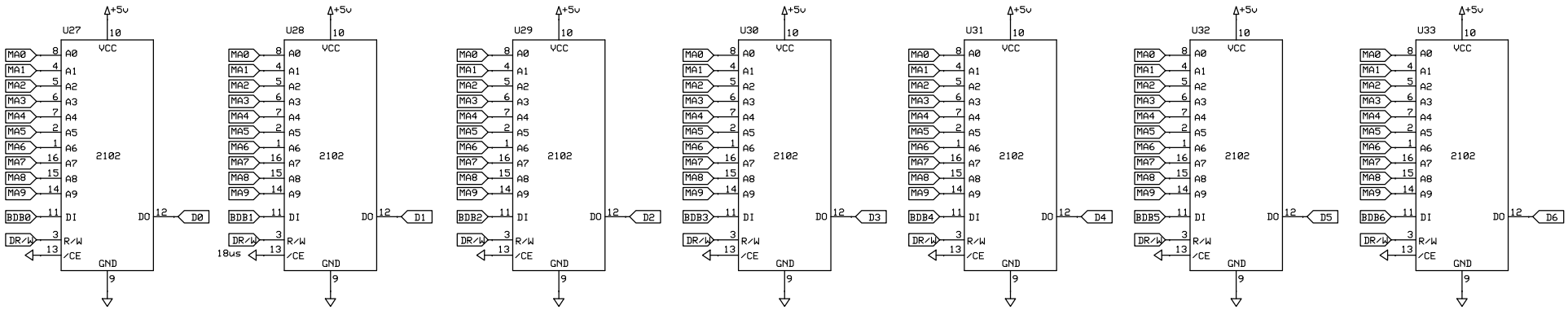
TMD Innovations		
TV Text Display		
Len Bayles	Rev 1.0 7/9/2015	Page # 1

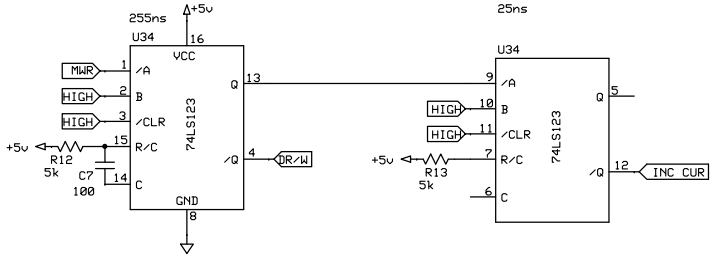


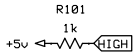
TMD Innovations		
TV Text Display		
Len Bayles	Rev 1.0 7/17/2015	Page # 2



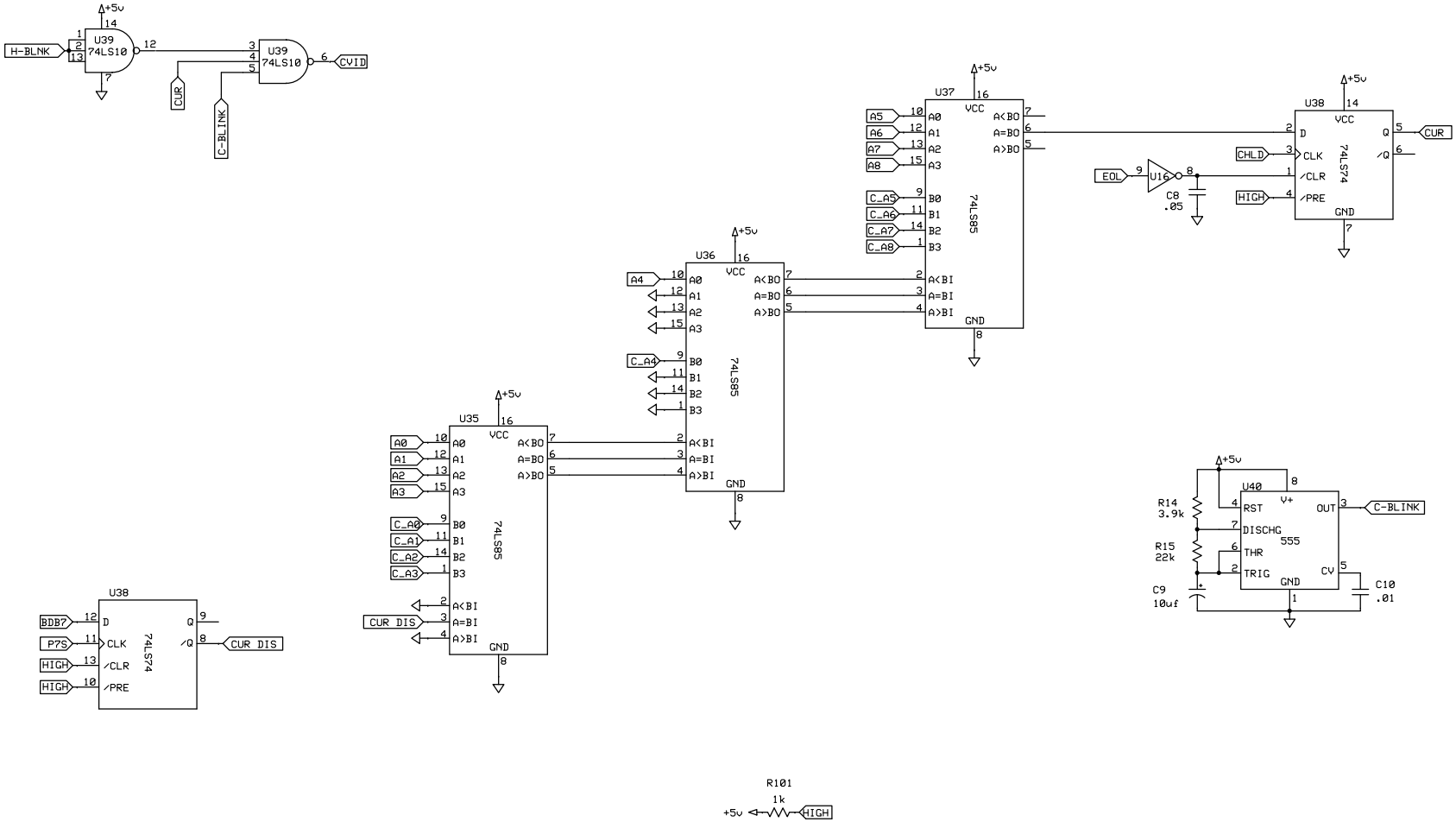
TMD Innovations		
TV Text Display		
Len Bayles	Rev 1.0 8/2/2015	Page # 3







TMD Innovations		
TV Text Display		
Len Bayles	Rev 1.0 7/7/2015	Page # 4



TMD Innovations		
TV Text Display		
Len Bayles	Rev 1.0 7/11/2015	Page # 5